WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

Memory

Lecture 19: registers, addressing, byte-addressable RAM and the register file

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Today

01 Registers

02 Addressing

03 Byte-addressable RAM

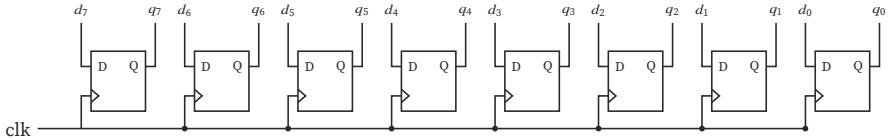
04 The register file

05 Summary

01

Registers

An 8-bit register

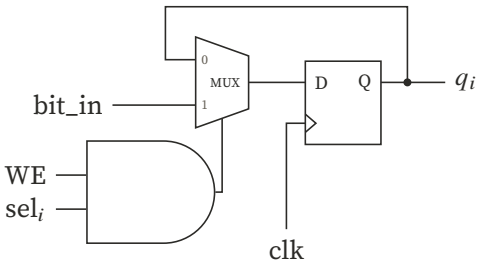


- Eight D flip-flops that share one clock: at each rising edge they store the 8 bits $d_7 \ldots d_0$ together. This is the book's 8-bit register.
- To store only when told, put a 2-to-1 multiplexer in front of each D `(Unit 12)`: it passes q_i back (keep) or d_i (load). Its select is the *write enable*.

02

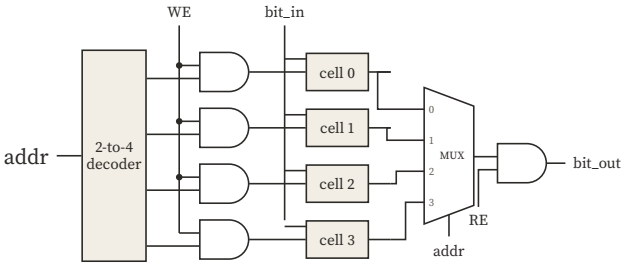
Addressing

One memory cell



- One flip-flop with a multiplexer in front: input 0 feeds q_i back, input 1 takes the new bit.
- It loads only when the memory is told to write (WE = 1) *and* this is the cell the address names ($\text{sel}_i = 1$).
- Every other cell keeps its value at the clock edge.

Four bits, one address



- The book's 4-bit memory: cells 00 to 11, one bit each.
- Write: the decoder (`Unit 8`) turns the address into one select line; with WE = 1 only that cell loads bit_in.
- Read: a 4-to-1 multiplexer (`Unit 7`) picks the addressed cell; with RE = 1 its bit appears at bit_out.

Write enable and read enable

- Every memory access names an *address* and says what to do with it:
  - **WE** (write enable): at the next clock edge, store the input at that address;
  - **RE** (read enable): put the content of that address on the output.

  They are also called *load* and *enable* bits.
- The book also routes bit_in to the addressed cell with a demultiplexer `(Unit 7)`. Since only that cell loads, the effect is the same.
- Reading takes no clock edge: the output follows the address, through the multiplexer. Writing waits for the edge, like every flip-flop.

03

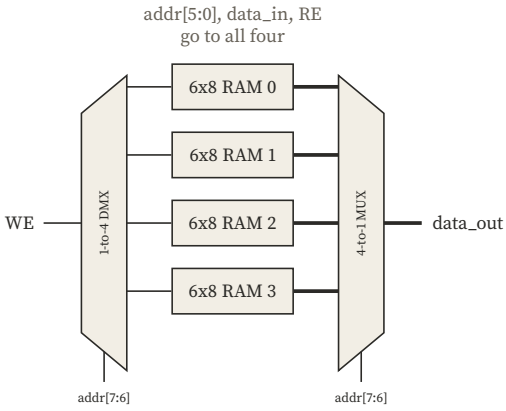
Byte-addressable RAM

From bits to bytes

- Put an 8-bit register in place of each flip-flop: the same circuit now stores 4 bytes. The multiplexers simply carry 8 bits instead of one.
- The book calls it a 2x8 RAM: 2 address bits, and 8 bits at each address. Each address names a byte: the memory is *byte-addressable* (Unit 15).

RAM	Address bits	Bytes	Built from
2x8	2	4	four 8-bit registers
4x8	4	16	four 2x8 RAMs
6x8	6	64	four 4x8 RAMs
8x8	8	256	four 6x8 RAMs

Four smaller RAMs make a bigger one



- The two top address bits choose one of four smaller RAMs: they steer WE to it, and pick its output.
- The rest of the address goes on down, into the chosen one.
- Repeat: each step adds 2 address bits and makes the memory 4 times bigger. The 8x8 RAM is the DX8's 256 bytes.

Random access

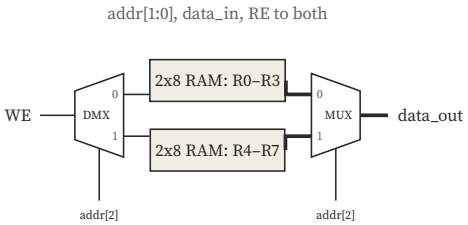
- *Random access*: any address can be read or written in the same time, in any order. Compare a tape, which must wind to the place it needs.
- The DX8's RAM holds $256 \times 8 = 2048$ bits: 2048 flip-flops, each with its multiplexer.
- Memory built from flip-flops is volatile: when the power goes off, its contents are lost.

> **Example**
>
> Address `0xC5` = 11 000101 in the 8x8 RAM: the top bits 11 choose 6x8 RAM 3, and inside it the address is 000101 = 5.

04

The register file

The register file



- The CPU's own small memory: 8 registers of 8 bits, R0 to R7, chosen by a 3-bit address. In sync with the clock, like the RAM.
- The book builds it as a 3x8 RAM, the same way: two 2x8 RAMs, with the third address bit choosing between them.
- An instruction names its register in $b_{10:8}$ `(Unit 15)`: that is the register file's address.

Memory in the DX8

	Size	Address	Holds
RAM (8x8)	256 bytes	8 bits, from the IP	program and data
Register file (3x8)	8 bytes	3 bits, from the instruction	R0 to R7
regA, regB, regOUT, …	1 byte each	none	one value each

- Who sets WE and RE, and when? The controller, step by step through every instruction (Unit 21).
- For example, the RAM is written only while a `store` instruction is in its last stage.

Try it

1. How many flip-flops are in a 4x8 RAM?
   16 bytes of 8 bits: 128.
2. How many address bits does a byte-addressable memory of 1024 bytes need?
   10, since $2^{10} = 1024$.
3. In the 4-bit memory, addr = 10, WE = 1, RE = 0, bit_in = 1. What happens at the clock edge, and what is bit_out?
   Cell 2 stores 1; the others keep their values. bit_out = 0, since RE = 0.
4. Which 6x8 RAM of the 8x8 RAM holds address `0x47`, and at which address inside it?
   $0x47 = 01\,000111$: RAM 1, address 7.
5. How many 2x8 RAMs does the 8x8 RAM contain?
   $4 \cdot 4 \cdot 4 = 64$ (and $64 \cdot 4 = 256$ bytes).

05

Summary

Summary

- A register is n flip-flops on one clock; a multiplexer in front of each D lets it keep its value unless written.
- An addressed memory: a decoder and WE choose the one cell that loads; a multiplexer and RE choose the one that is read.
- With bytes in place of bits, 2^n addresses give a byte-addressable nx8 RAM. Four smaller RAMs and two more address bits make a RAM four times bigger.
- The DX8 has a 256-byte RAM (8x8) and a register file of 8 registers (3x8), built the same way.

Before the next lecture

- Read Section 6.5 of the book, and the register file in Section 6.6.2.
- In Logisim-evolution, build the 4-bit memory, then the 2x8 RAM. Build the register file (3x8) from two 2x8 RAMs.
- Read the beginning of Section 6.6 and Section 6.6.1.

> **Next lecture** (Unit 20)
>
> Processor design: fetch, decode, execute and load; single-cycle, multi-cycle and pipelined designs.